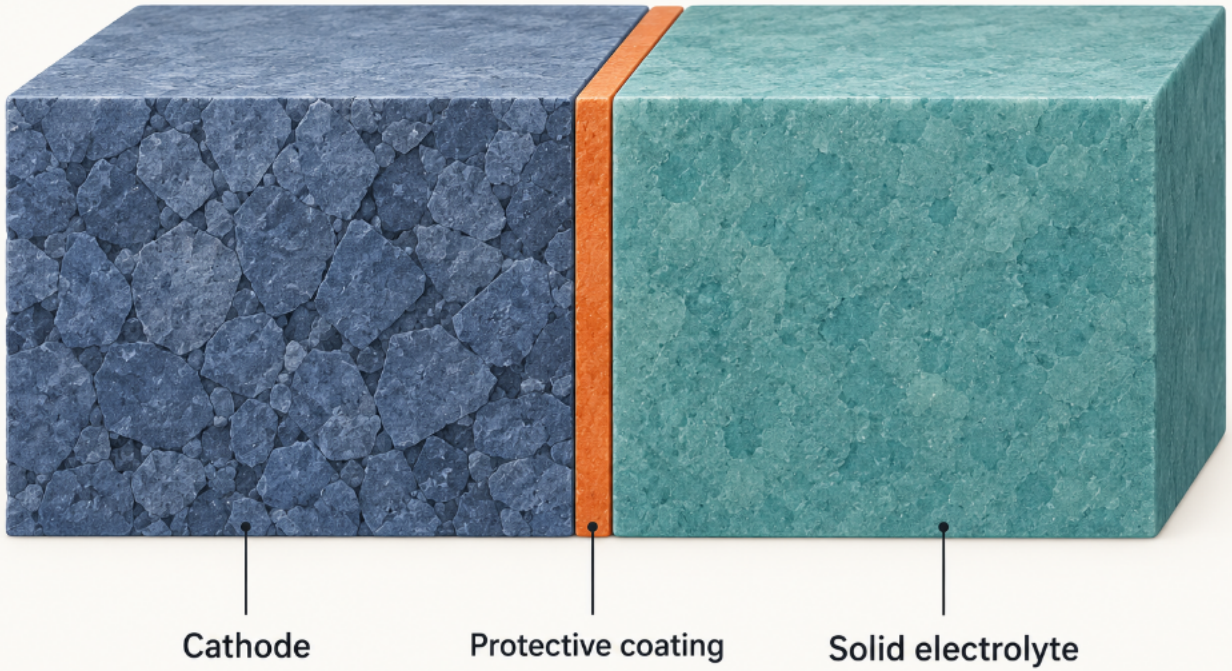
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interphAIse: choosing protective battery layers

An AI language model chooses what to test. Scientific tools check its choices.



Conceptual illustration, not to scale

A coating must

Limit unwanted reactions and stay attached.

Block electrons.
Let lithium ions cross.

Goal: test whether feedback improves coating selection.

Davide Tisi and Adem R. N. Aouichaoui

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Our objective: better choices within a fixed budget

Can calculation feedback help an LLM find acceptable coatings?

Scientific goal

Find protective coatings that are compatible, attached and able to carry lithium ions.

Testable question

Does using the true result improve the next material choice, compared with the controls?

Measured outcome

Find at least one acceptable candidate within ten choices.

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Four requirements for a useful coating

Chemical acceptance is one part of the physical problem.

Requirement	What it means	What this study checks
Compatibility	Limit decomposition and reactions	Fixed reaction-score gate
Electronic insulation	Block electronic leakage	Band-gap gate
Adhesion	Keep the layers attached	Three coatings compared
Ion transport	Let lithium ions cross	Not yet tested

The search screen checks the first two. A useful coating needs all four.

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Why the LLM is central

The question is which material deserves the next test.

Propose

Connect chemical knowledge to a candidate and explain why it is worth testing.

Use the result

Read the numerical feedback and choose where to spend the next query.

Make a forecast

Predict a physical result before the calculation so the explanation can be challenged.

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The LLM and its scientific tools

Choose a candidate, calculate its properties, read the result, choose again.

Part of the workflow	Tool and role
Scientific choices	Haiku 5.5 / DeepSeek V3.2 Select a known material and explain the choice
Chemical screening	Materials Project + pymatgen / SciPy Supply reference data and precompute reaction scores
Physical follow-up	ASE + PET-MAD Model the atoms and calculate attachment

141 known crystal structures representing 100 compositions.

Three electrolytes. Ten choices per search. 810 completed searches.

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Why these battery materials?

Keep $LiCoO_2$ at 4 V fixed so the electrolyte chemistry is the variable.

Electrolyte	Why it was included
Li_6PS_5Cl / sulfide	Established reference for the coating search
Li_3InCl_6 / chloride	Tests a different anion chemistry An ordered structure was available in the cache
LLZO / oxide	Tests whether a useful chemical direction still works in an oxide environment

Three contrasting chemistries test how broadly the selection strategy works.

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Our screening definition of a good coating

A candidate must pass both gates. Different runs may find the same material.

Requirement	Acceptance rule
Chemical compatibility	Penalty ≤ 0.050 eV per non-Li atom
Electronic insulation	Band gap ≥ 2 eV

Penalty uses the worst of coating self-decomposition and the extra reaction at either contact.

Contact reactions use separately equilibrated reference materials.

Search success: at least one passing candidate within ten choices.

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Five strategies test the value of feedback

Same menu, ten choices and 30 searches per strategy.

Strategy	How it chooses
Uses results	One choice at a time, with the true result after each test
No results	One at a time, remembering choices but seeing no scores
Mixed-up results	One at a time, with scores assigned to the wrong materials
One list	All ten choices together, before seeing any result
Random	Software samples ten materials without repeats

Success = finding at least one material that passes the fixed chemical screen.

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What each comparison can tell us

The strongest feedback tests keep the one-at-a-time choice policy.

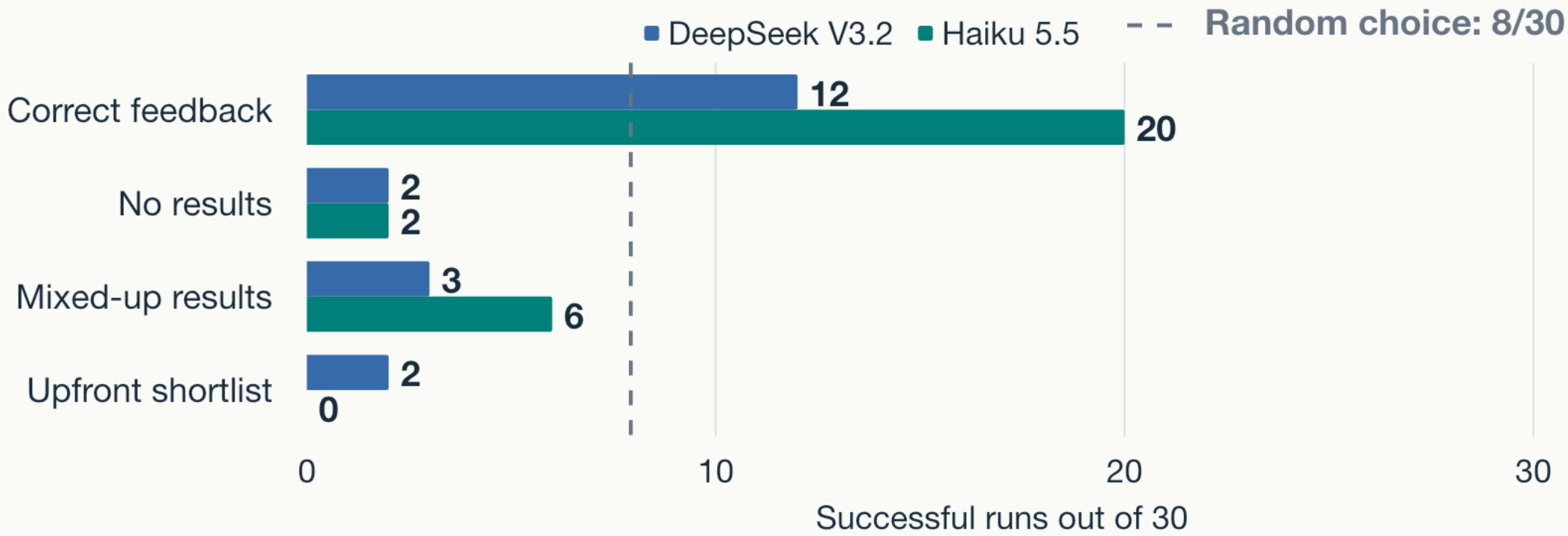
Compare true results with	Question answered
No results	Does seeing numerical outcomes help?
Mixed-up results	Does correct material–result attribution matter?
One upfront list	How useful is initial candidate selection?
Random	How does the observed success compare with simple sampling?

Upfront also changes the choice policy. It cannot isolate feedback alone.

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Sulfide: correct feedback helps Haiku

Success = at least one accepted candidate within ten choices. Each strategy has 30 runs.



Correct feedback: Haiku 20 and DeepSeek 12, compared with random 8.

Haiku beats both controls after correction. DeepSeek remains inconclusive.
Cross-model contrasts are descriptive. Reasoning compute differs.

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A real example: refining a near miss

Three consecutive choices in one sulfide search. Lower reaction score is better.

Try	Material chosen	Reaction score	What the test said
6	Li_3AlF_6	0.160	Does not pass
7	$LiMgAlF_6$	0.057	Close, but still above the limit
8	$LiCaAlF_6$	0.048	Passes the 0.050 limit

The LLM kept testing related chemistry after a promising near miss.

One illustrative search. Reaction scores are in eV per non-lithium atom.

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Why these 141 coating candidates?

A broad, reproducible menu of known lithium-containing materials.

Selection rule	Reason
Known Materials Project structures	Start from reproducible atomic models
Near-stable, ordered, small cells	Keep the chemistry and calculation size manageable
All eligible records retained	Avoid choosing the menu using the new results

Oxides, halides, silicates, phosphates and borates provide chemical variety.

The sulfide screen accepts four structures: $LiNbF_6$, two $LiTaF_6$ and $LiCaAlF_6$.

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Why these three coatings were calculated

All three pass the sulfide chemical screen and have feasible interface models.

Coating	Reason for the choice
$LiNbF_6$ / reference	Already calculated and chemically acceptable Gives Haiku a physical reference
$LiTaF_6$ / first choice	Acceptable and insulating Haiku expects similar behavior to the Nb analogue
$LiCaAlF_6$ / second choice	Only acceptable unevaluated option left Its Ca/Al chemistry is a less-close analogy

The reasons are recorded before testing. The forecasts can still fail.

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Haiku's forecasts were recorded before the tests

PET-MAD results in J/m². Forecast ranges are model statements, not confidence intervals.

Material / contact	Haiku forecast	Calculated	Outcome
$LiTaF_6$ / cathode	3.2–3.9	3.339	Within range
$LiTaF_6$ / electrolyte	1.4–1.9	1.577	Within range
$LiCaAlF_6$ / cathode	3.0–3.8	1.427	Below range
$LiCaAlF_6$ / electrolyte	1.4–1.9	0.744	Below range

The electrolyte is the weaker contact in every case, as Haiku predicted.

Choice order stayed unchanged. Two new choices cannot establish reliable adaptive search.

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Sulfide follow-up: a small candidate shortlist

Three coatings tested with PET-MAD. Higher adhesion means stronger attachment.

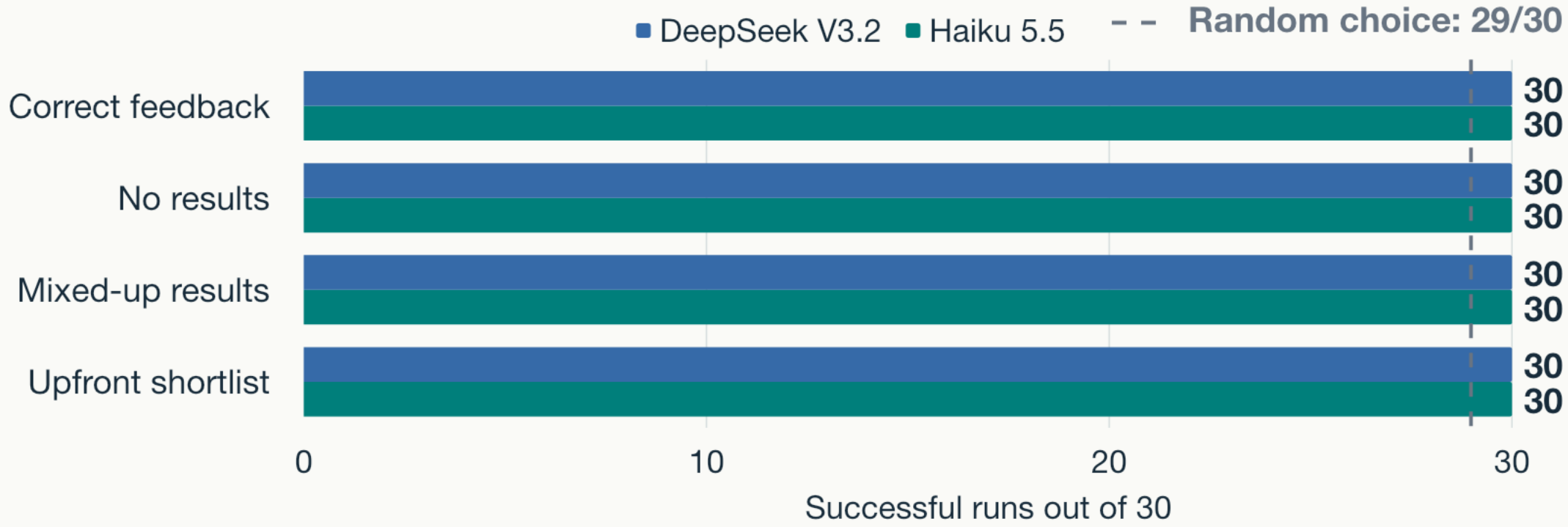
Coating	Weaker-contact adhesion (J/m²)	Haiku's forecasts
$LiNbF_6$	1.633	Reference supplied to Haiku
$LiTaF_6$	1.577	Both within range
$LiCaAlF_6$	0.744	Both too high

$LiNbF_6$ and $LiTaF_6$ remain a follow-up pair, without a reliable winner.

Provisional values for formally polar interfaces. Lithium transport is untested.
Nb–Ta gap: 0.056 J/m², below the ~0.12 J/m² bulk-reference sensitivity.

Chloride: almost every strategy succeeds

Success = at least one accepted candidate within ten choices. Each strategy has 30 runs.



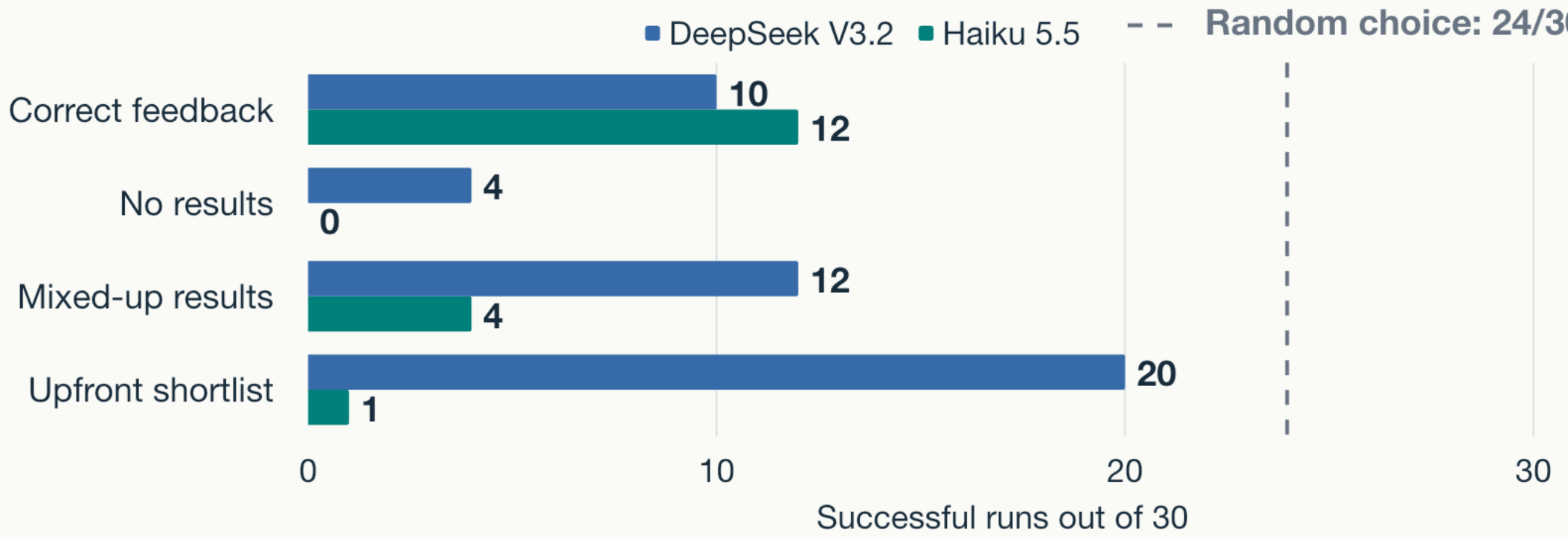
All LLM strategies: 30/30. Random choice: 29/30.

55 of 141 structures pass. This easy screen leaves little room for a feedback advantage.

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LLZO: the feedback loop falls short

Success = at least one accepted candidate within ten choices. Each strategy has 30 runs.



Every LLM strategy trails random choice on the observed success count.

DeepSeek: 20 upfront, 10 with feedback. This also changes the selection policy.
The acceptance proxy admits questionable oxyhalogens. The frozen counts stay unchanged.

LLZO separates starting knowledge from feedback use

DeepSeek can propose useful candidates without improving through this loop.

Useful starting list

20/30 upfront searches succeed.
They include $LiAlSiO_4$ and $LiMgPO_4$.

Weak feedback use

10/30 with true results, 12/30 with mixed-up results.
No clear benefit in the planned comparisons.

Meaning of the failure

The tested strategy needs improvement.
The upfront comparison changes more than feedback.

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What interphAIse has demonstrated

Scientific choices can be tested against controlled feedback and physical outcomes.

Feedback can help

Haiku benefits clearly in sulfide.
The benefit depends on the environment and model.

Forecasts can fail

Ta forecasts held; Ca–Al forecasts were too high.
A plausible explanation must face a calculation.

A shortlist remains

$LiNbF_6$ and $LiTaF_6$ deserve further testing.
A validated, ion-conducting winner is still open.

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What the LLM contributes

Useful chemical knowledge and useful adaptation to feedback are distinct capabilities.

Next: test lithium transport through the shortlisted coatings.

Then validate with higher-accuracy calculations and experiments.
No human-expert comparison or validated coating yet.

Code and evidence: github.com/DavideTisi/interphAIse

Structure viewer: davidetisi.github.io/interphAIse

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Complete production results and uncertainty

Successes / 30 with 95% Wilson intervals below. Random is shared across models.

Electrolyte	Model	Uses results	No results	Mixed-up results	One list	Shared random
Sulfide	Haiku	20/30 49–81%	2/30 2–21%	6/30 10–37%	0/30 0–11%	8/30 14–44%
Sulfide	DeepSeek	12/30 25–58%	2/30 2–21%	3/30 3–26%	2/30 2–21%	8/30 14–44%
Chloride	Haiku	30/30 89–100%	30/30 89–100%	30/30 89–100%	30/30 89–100%	29/30 83–99%
Chloride	DeepSeek	30/30 89–100%	30/30 89–100%	30/30 89–100%	30/30 89–100%	29/30 83–99%
LLZO	Haiku	12/30 25–58%	0/30 0–11%	4/30 5–30%	1/30 1–17%	24/30 63–90%
LLZO	DeepSeek	10/30 19–51%	4/30 5–30%	12/30 25–58%	20/30 49–81%	24/30 63–90%

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Planned tests distinguish a trend from evidence

Correct feedback versus each control. Exact paired tests with Holm correction.

Electrolyte	Compared with	Haiku adjusted p	DeepSeek adjusted p
Sulfide	No results	0.00004578	0.07764
Sulfide	Mixed-up results	0.00244	0.11230
Chloride	No results	1	1
Chloride	Mixed-up results	1	1
LLZO	No results	0.00244	0.43750
LLZO	Mixed-up results	0.17212	1

Three Haiku contrasts pass 0.05. None of DeepSeek's six does.

Separate test families. This is not a statistical test of one model against the other.

LLZO: practical annotations do not replace the score

Unflagged means outside one defined oxyhalogen review set, not a validated battery coating.

Policy	All successes	Unflagged-hit runs	Flagged-only runs
Shared random	24	18	6
Haiku, uses results	12	12	0
DeepSeek, one list	20	20	0
DeepSeek, uses results	10	9	1

DeepSeek's upfront list found $LiAlSiO_4$, $LiMgPO_4$ and $LiAl(SiO_3)_2$.

22 acceptable picks across 20 successful runs. The analysis is descriptive and post-hoc.

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Relaxed adhesion and rigid separation are different

Coating	Cathode Wadh / Wsep	Electrolyte Wadh / Wsep	Atoms C / E
$LiNbF_6$	3.617 / 5.168	1.633 / 2.967	696 / 880
$LiTaF_6$	3.339 / 4.728	1.577 / 2.911	696 / 880
$LiCaAlF_6$	1.427 / 4.322	0.744 / 1.490	720 / 1212

All values in J/m². Wadh allows separated slabs to relax.
Wsep keeps their joined geometry. Both contacts are formally polar.
LiF remains uncomputed.

Ca–Al electrolyte thickness and bulk-structure sensitivity remain unquantified.
Primary snapshots unchanged. Later robustness checks are separate.